# MATH 3042- Assignment 2- Winter 2026- Solutions

January 27, 2026 12:24 PM

**Assignment 2: Summarizing Data - Numerical Methods**

**Course:** MATH 3042: Applied Probability and Statistics for CST

**Due Date:** Jan 23rd by 12:30 pm

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Full Name:

Set:

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**Instructions**

- Complete your assignment in this Word document
- Manual calculations help you develop statistical intuition, catch errors when results seem wrong, and verify that your code is working correctly. In this assignment, please show all steps through manual calculations (calculator allowed) without using R.
- Include all answers, explanations, and interpretations in this document
- **Convert your final Word document to PDF before submission**

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***Q1. (8 marks)*** You have response times (in ms) from two API endpoints:

Endpoint A: [45, 50, 52, 47, 49, 51, 50, 48]
Endpoint B: [30, 35, 250, 33, 30, 300, 35]

By showing all steps, calculate mean, median, standard deviation, and IQR for both. Which endpoint is more reliable? Why?

| locator | A | B |
|---|---|---|
| 1 | 45 | 30 |
| 2 | 47 | 30 |
| 3 | 48 | 33 |
| 4 | 49 | 35 |
| 5 | 50 | 35 |
| 6 | 50 | 250 |
| 7 | 51 | 300 |
| 8 | 52 | |
| | | |
| Sum | 392 | 713 |
| Mean | 49 | 101.9 |

Mean for set A: $\frac{392}{8} = 49$
Mean for set B: $\frac{713}{7} = 101.9$

Median for set A: $l = 8 \times \frac{50}{100} = 4 \quad Median = \frac{49+50}{2} = 49.5$
Median for set B: $l = 7 \times \frac{50}{100} = 3.5\ round\ up: 4 \quad Median = 35$

Q1 for set A: $l = 8 \times \frac{25}{100} = 2 \quad Q_1 = \frac{47+48}{2} = 47.5$
Q1 for set B: $l = 7 \times \frac{25}{100} = 1.75\ round\ up: 2 \quad Q_1 = 30$

Q3 for set A: $l = 8 \times \frac{75}{100} = 6 \quad Q_3 = \frac{50+51}{2} = 50.5$
Q3 for set B: $l = 7 \times \frac{75}{100} = 5.25 \quad round\ up\ 6: Q_3 = 250$

IQR for set A: $50.5 - 47.5 = 3$
IQR for set B: $250 - 30 = 220$

| | A: (x-xbar)^2 | B: (x-xbar)^2 |
|---|---|---|
| | 16 | 5163.45 |
| | 4 | 5163.45 |
| | 1 | 4741.31 |
| | 0 | 4469.88 |
| | 1 | 4469.88 |
| | 1 | 21946.31 |
| | 4 | 39260.59 |
| | 9 | |
| | | |
| sum | 36 | 85214.86 |
| Var | 5.1 | 14202.5 |
| Stdev | 2.3 | 119.2 |

$$set\ A\ var = \frac{36}{7} = 5.1 \qquad stdev = \sqrt{5.1} = 2.3$$

$$set\ B\ var = \frac{85214.86}{6} = 14202.5 \qquad stdev = \sqrt{14202.5} = 119.2$$

The endpoint A is more reliable since it has a much lower IQR as well as a lower standard deviation

***Q2. (7 marks)*** Daily active users over 14 days: [1200, 1250, 1180, 1300, 1220, 15000, 1190, 1240, 1210, 1230, 1200, 1180, 1260, 1220]

Identify outliers using the IQR method. Show all steps.

| locator | Active users sorted |
|---|---|
| 1 | 1180 |
| 2 | 1180 |
| 3 | 1190 |
| 4 | 1200 |
| 5 | 1200 |
| 6 | 1210 |
| 7 | 1220 |
| 8 | 1220 |
| 9 | 1230 |
| 10 | 1240 |
| 11 | 1250 |
| 12 | 1260 |
| 13 | 1300 |
| 14 | 15000 |

$$Q_1: l = 14 \times \frac{25}{100} = 3.5\ round\ up: 4 \qquad Q_1 = 1200$$

$$Q_3: l = 14 \times \frac{75}{100} = 10.5\ round\ up: 11 \quad Q_3 = 1250$$

$$IQR = 1250 - 1200 = 50$$

$$UF = 1250 + 1.5 \times 50 = 1325$$
$$LF = 1200 - 1.5 \times 50 = 1125$$

There are no values less than 1125, we do not have lower outliers
The is one value (15000) above 1350, we have one upper outlier.

***Q3. (5 marks)*** Server CPU usage percentages: [ 47, 49, 95, 46, 50, 48]

Calculate all z-scores and identify anomalies (|z| > 2).

| | CPU Usage | Z-scores |
|---|---|---|
| | 47 | -0.459 |
| | 49 | -0.355 |
| | 95 | 2.036 |
| | 46 | -0.511 |
| | 50 | -0.303 |
| | 48 | -0.407 |
| | | |
| Mean | 55.8 | |
| Stdev | 19.2 | |

There is one slightly anomalous value: 95

***Q4. (5 marks)*** Your CDN has these response times (in milliseconds) for 20 requests:

**Dataset:** 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 45, 50, 55, 60, 70, 85, 100, 200

If you want to guarantee that 95% of users experience sub-100ms latency, are you meeting that goal?

$n = 20$
$number\ of\ responses\ less\ that\ 100ms: 18$
$Percent\ of\ responses\ less\ than\ 100\ ms: \frac{18}{20} = 90\%$
No, we are not meeting that goal.